AI SALES AGENT

Functional requirements

Observable Phase 1 behaviour, identified and prioritised from the product requirements register.

Phase 1 MVP · 21 October 2026 · Prepared from the canonical documents in docs/

How to read this register

Identifiers, statements, and priorities are transcribed from `docs/00_product/01_PRODUCT_REQUIREMENTS`. MoSCoW applies to Phase 1. **Must** is required for the MVP. **Should** is in the baseline but may slip. **Could** is optional. **Won't** is explicitly out of this phase. Verification codes are AT (automated test), MT (manual test), EV (AI evaluation), OB (production observation), and IN (inspection).

> Where a later engineering document is more specific, that document wins for detail. This register wins for the observable behaviour. Conflicts already resolved in `41_ASSUMPTIONS.md` are followed here, including the 30-day auto-close and suggestions only on an explicit request.

Counts: 222 Must, 21 Should, 3 Could, 17 Won't, 274 requirements in total.

4. FR-AUTH — Authentication and identity

ID	Observable behaviour	Priority	Verify
FR-AUTH-001	A visitor can register with email and password, creating a `users` row.	**Must**	AT
FR-AUTH-002	Passwords are stored only as a salted hash; plaintext is never logged, returned, or persisted.	**Must**	AT, IN
FR-AUTH-003	A registered user can log in and receive a short-lived access token plus a refresh token.	**Must**	AT
FR-AUTH-004	A user can refresh an expired access token without re-entering credentials, until the refresh token expires or is revoked.	**Must**	AT
FR-AUTH-005	A user can log out, invalidating the refresh token for that session.	**Must**	AT
FR-AUTH-006	A user can sign in with an OAuth provider (Google) and be linked to an existing `users` row by verified email.	**Should**	AT
FR-AUTH-007	A user belonging to multiple businesses can select the active business; every API call resolves tenant scope from the token or an explicit business header.	**Must**	AT
FR-AUTH-008	A user can request a password reset by email and set a new password with a single-use, time-limited token.	**Should**	AT
FR-AUTH-009	Failed login attempts are rate-limited per email and per IP.	**Must**	AT, OB
FR-AUTH-010	Email addresses are unique and case-insensitive across the platform.	**Must**	AT
FR-AUTH-011	An invited user can accept an invitation and join an existing business with the role assigned by the inviter.	**Must**	AT
FR-AUTH-012	Multi-factor authentication.	**Won't**	—
FR-AUTH-013	A public landing page at `/` presents the commercial value proposition, bilingual (Derja/French) sales agent preview, 4-KPI ROI, and navigation CTAs to `/login` and `/register`.	**Must**	AT, MT

5. FR-BIZ — Business workspace, members, onboarding

ID	Observable behaviour	Priority	Verify
FR-BIZ-001	An authenticated user can create a business, becoming its Owner.	**Must**	AT
FR-BIZ-002	A business captures name, industry (default `fashion`), country (default `TN`), default language (`fr` or `ar`), timezone, and currency (default `TND`).	**Must**	AT
FR-BIZ-003	Creating a business seeds: one default `ai_agent`, default `ai_settings`, platform default `ai_prompts` references, and a starter `business_rules` set (delivery, payment, returns) marked as unconfirmed defaults.	**Must**	AT
FR-BIZ-004	Owner and Admin can update business profile fields.	**Must**	AT
FR-BIZ-005	Only the Owner can delete a business; deletion is soft (30-day recovery window) and immediately disables AI and channel processing.	**Should**	AT
FR-BIZ-006	All tenant data access is scoped by `business_id`; no endpoint may return data belonging to another business.	**Must**	AT, IN
FR-BIZ-007	Ownership transfer between users.	**Won't**	—
FR-BIZ-010	Owner and Admin can invite a user by email with a role of Admin or Employee (merchant UI may label Employee as **Agent**).	**Must**	AT
FR-BIZ-011	Owner and Admin can remove a member; removal revokes access immediately and releases that user's active conversation assignments.	**Must**	AT
FR-BIZ-012	Owner and Admin can change a member's role, except that only the Owner may grant Owner-equivalent capability (Phase 1: no second Owner).	**Must**	AT
FR-BIZ-013	The member list shows name, email, role, invite status, and last active timestamp.	**Should**	MT
FR-BIZ-014	Seat count is enforced against the plan limit; inviting beyond the limit returns a quota error with an upgrade CTA.	**Must**	AT
FR-BIZ-015	Custom roles with granular permission selection.	**Won't**	—
FR-BIZ-020	The system tracks per-step completion flags for: business created, channel connected, products imported, AI personality + minimum policies configured, AI playground test passed, handoff workflow acknowledged, AI activated.	**Must**	AT
FR-BIZ-021	The user can leave and resume onboarding at any point without data loss; the checklist state is server-side, not client-side.	**Must**	AT
FR-BIZ-022	The user can skip optional steps and return to them from Settings.	**Must**	MT
FR-BIZ-023	The checklist displays remaining blocking items and links directly to each.	**Must**	MT
FR-BIZ-024	The system computes `ai_activation_ready` from the gates in §14 and exposes it to the UI.	**Must**	AT
FR-BIZ-025	Attempting to set AI live while `ai_activation_ready` is false is rejected by the API, not only by the UI.	**Must**	AT
FR-BIZ-026	An AI Playground lets Owner/Admin simulate conversations (price question, angry customer, buying intent, product recommendation) without a connected channel; passing it satisfies the "AI test" gate.	**Must**	AT, MT

6. FR-CAT — Product catalog

ID	Observable behaviour	Priority	Verify
FR-CAT-001	Owner/Admin can create, read, update, and archive products with name, description, category, status (`draft` / `active`), and currency override.	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-CAT-002	Only `active` products with at least one in-stock variant are visible to AI product search.	**Must**	AT, EV
FR-CAT-003	Owner/Admin can manage a category tree (`parent_id`, `sort_order`) scoped to the business.	**Must**	AT
FR-CAT-004	Owner/Admin can create variants with SKU, barcode, attributes (JSONB, e.g. `{"color":"Black","size":"M"}`), price, optional discount price, and stock.	**Must**	AT
FR-CAT-005	SKU is unique per business when provided; blank SKUs are permitted and auto-generated.	**Must**	AT
FR-CAT-006	Price must be non-negative; discount price, when set, must be less than price.	**Must**	AT
FR-CAT-007	Stock is a non-negative integer; the system exposes `available_stock = stock - active reservations`.	**Must**	AT
FR-CAT-008	Owner/Admin can upload images to a product or to a specific variant, with alt text, sort order, and one primary flag per scope.	**Must**	AT
FR-CAT-009	Images are stored in object storage; the database stores URLs, never binary blobs.	**Must**	IN
FR-CAT-010	Optional AI merchandising fields (`benefits`, `selling_points`, `objections`, `faqs`) can be set per product and are included in retrieval context.	**Should**	AT, EV
FR-CAT-011	Editing price, stock, name, description, or merchandising fields enqueues a re-embedding job for the affected records.	**Must**	AT
FR-CAT-012	Deleting a product that appears on an order is blocked; the product must be archived instead.	**Must**	AT
FR-CAT-013	`search_products` (and equivalent retrieval) returns sellable **variants** (SKU, size, color, price, available stock), never an unpriced parent row.	**Must**	AT, EV
FR-CAT-015	Bulk edit of price or stock across selected variants.	**Could**	AT
FR-CAT-014	Inventory movement history (`inventory_movements`) recording every stock delta with a reason.	**Should**	AT
FR-CAT-020	Owner/Admin can upload a CSV of products; required columns are name, price, stock; SKU, category, attributes, and image URL are optional.	**Must**	AT
FR-CAT-021	The system validates the file and returns a **row-level error report** before any data is written.	**Must**	AT
FR-CAT-022	The user previews the parsed result and explicitly confirms before import executes.	**Must**	MT
FR-CAT-023	Confirmed import runs asynchronously and upserts categories, products, variants, and image rows.	**Must**	AT
FR-CAT-024	Import runs are recorded with status, row counts, and a downloadable error file.	**Must**	AT
FR-CAT-025	Successful import enqueues embedding reindex for affected records and marks the catalog onboarding step complete.	**Must**	AT
FR-CAT-026	Re-importing the same file is idempotent by SKU: existing variants update rather than duplicate.	**Must**	AT
FR-CAT-027	A downloadable CSV template with correct headers and one example row is provided.	**Must**	MT

ID	Observable behaviour	Priority	Verify
FR-CAT-028	Google Sheets import; Shopify / WooCommerce sync.	**Won't**	— (Phase 2, see 04_ROADMAP.md)

7. FR-KNOW — Knowledge base and business rules

ID	Observable behaviour	Priority	Verify
FR-KNOW-001	Owner/Admin can define delivery rules: zones with fee and estimated duration.	**Must**	AT
FR-KNOW-002	Owner/Admin can define accepted payment methods (e.g. cash on delivery, bank transfer).	**Must**	AT
FR-KNOW-003	Owner/Admin can define the return/exchange policy: window in days and conditions.	**Must**	AT
FR-KNOW-004	Owner/Admin can define working hours, used when the AI promises human follow-up timing.	**Must**	AT
FR-KNOW-005	Owner/Admin can define discount policy; the default is that the AI may offer **no** discounts.	**Must**	AT, EV
FR-KNOW-006	Owner/Admin can define minimum order value and free-shipping threshold.	**Should**	AT
FR-KNOW-007	Rules carry `priority`, `enabled`, `effective_from`, and `effective_to`; the AI resolves rules effective at the conversation timestamp.	**Must**	AT
FR-KNOW-008	Editing a rule creates a new effective version rather than destroying history.	**Must**	AT
FR-KNOW-009	Delivery fees and payment answers are computed from `business_rules`, never inferred by the model.	**Must**	AT, EV
FR-KNOW-010	When two enabled rules conflict for the same question, the system does not guess: it escalates with reason `policy_conflict`.	**Must**	AT, EV
FR-KNOW-011	Owner/Admin can create, edit, and delete knowledge documents (title, source type, raw text).	**Must**	AT
FR-KNOW-012	Saving a document chunks and embeds it asynchronously; the UI shows indexing status.	**Must**	AT
FR-KNOW-013	Re-saving unchanged content does not re-embed (content-hash check).	**Should**	AT
FR-KNOW-014	Owner/Admin can trigger a full reindex for the business.	**Must**	AT
FR-KNOW-015	File upload (PDF/DOCX) knowledge ingestion.	**Could**	AT
FR-KNOW-016	Website crawl ingestion.	**Won't**	—

8. FR-INT — Channel integrations

ID	Observable behaviour	Priority	Verify
FR-INT-001	Owner/Admin can start a Meta OAuth flow from the Integrations page.	**Must**	MT
FR-INT-002	After OAuth, the user selects which Facebook Page(s) to connect; a linked Instagram professional account is connected with the Page when available.	**Must**	MT
FR-INT-003	Each connected surface is stored as a `channel_account` with channel type, external account id, credentials reference, status, and webhook secret.	**Must**	AT
FR-INT-004	Provider credentials are stored encrypted; raw tokens are never returned by any API or written to logs.	**Must**	AT, IN
FR-INT-005	The integrations page shows per-account connection status, last inbound message time, and last error.	**Must**	MT

ID	Observable behaviour	Priority	Verify
FR-INT-006	Owner/Admin can disconnect a channel account; disconnection stops inbound processing and outbound sends but preserves history.	**Must**	AT
FR-INT-007	Inbound webhooks are signature-verified; invalid signatures are rejected and logged.	**Must**	AT
FR-INT-008	The webhook endpoint persists the raw payload and returns 200 before any AI processing occurs.	**Must**	AT, OB
FR-INT-009	Duplicate webhook deliveries produce no duplicate message, no duplicate AI run, and no duplicate order.	**Must**	AT
FR-INT-010	Provider payloads are mapped to the canonical internal message shape; no provider-specific field reaches the AI or inbox modules.	**Must**	IN
FR-INT-011	Outbound sends respect provider rate limits with exponential backoff and a dead-letter queue after exhausting retries.	**Must**	AT, OB
FR-INT-012	Token expiry or permission revocation sets the channel account to an error state and notifies Owner/Admin.	**Must**	AT
FR-INT-013	Inbound attachments (images) are stored and shown in the inbox even though MVP AI does not interpret image content.	**Must**	AT
FR-INT-014	WhatsApp channel (Meta WhatsApp Cloud API).	**Won't**	— (Phase 2, behind feature_flags.whatsapp_enabled)
FR-INT-020	Connect a WhatsApp Business number via Meta OAuth; store WABA id and phone number id.	**S (Phase 2)**	MT
FR-INT-021	WhatsApp prerequisites stated before the merchant starts the flow.	**S (Phase 2)**	MT
FR-INT-022	Pending WhatsApp verification never blocks MVP channels or AI activation.	**S (Phase 2)**	AT
FR-INT-023	Each inbound WhatsApp message opens or refreshes a 24-hour service window.	**S (Phase 2)**	AT
FR-INT-024	Inside the service window, AI and human replies send as normal free-form messages.	**S (Phase 2)**	AT
FR-INT-025	Outside the service window, no autonomous AI reply and no free-form send attempt.	**S (Phase 2)**	AT
FR-INT-026	Outside the window, thread marked closed; staff may send an approved template.	**S (Phase 2)**	MT
FR-INT-027	Templates are pre-registered with Meta; never LLM-generated.	**S (Phase 2)**	AT, IN
FR-INT-028	Approved templates synced from Meta with approval status.	**S (Phase 2)**	AT
FR-INT-029	Provider-refused sends never reported as delivered.	**S (Phase 2)**	AT
FR-INT-030	WhatsApp conversation starts metered separately (Meta per-conversation cost).	**S (Phase 2)**	AT
FR-INT-031	Per-channel-account outbound throttling respects each provider's limits.	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-INT-032	Unofficial WhatsApp automation prohibited in the codebase.	**Must**	IN
FR-INT-033	Interactive WhatsApp types and in-app template authoring.	**Won't**	— (Phase 3)

9. FR-CONV — Conversations and inbox

ID	Observable behaviour	Priority	Verify
FR-CONV-001	Staff see a unified inbox of all conversations for the active business across all connected channels.	**Must**	MT
FR-CONV-002	The inbox can be filtered by status, channel, and assignee.	**Must**	AT
FR-CONV-003	Conversations requiring human attention are sorted first by handoff priority (`critical` → `high` → `normal`), then by handoff time (oldest first).	**Must**	AT
FR-CONV-004	Each list row shows customer name/handle, channel, last message preview, unread state, status badge, priority badge, and assignee.	**Must**	MT
FR-CONV-005	New inbound messages and status changes appear in an open inbox without manual refresh.	**Must**	MT
FR-CONV-006	The inbox supports search by customer name, phone, or message text.	**Should**	AT
FR-CONV-007	Message history is paginated and loads older messages on demand.	**Must**	AT
FR-CONV-008	Every message displays its sender type (customer, AI, human, system) with visually distinct treatment.	**Must**	MT
FR-CONV-010	The detail view shows the full transcript, the customer profile, lifecycle stage, order timeline, AI status, and the handoff briefing when present.	**Must**	MT
FR-CONV-011	Staff can send a message to the customer; sending as a human sets `ai_enabled = false` for that conversation.	**Must**	AT
FR-CONV-012	Staff can explicitly **Take over**, which pauses the AI and transitions `HUMAN_REQUIRED` → `HUMAN_HANDLING`.	**Must**	AT
FR-CONV-013	Staff can **Resume AI**, permitted only when no order is awaiting confirmation on that conversation.	**Must**	AT
FR-CONV-014	Staff can assign a conversation to a member; assignment history is recorded with actor and timestamp.	**Must**	AT
FR-CONV-015	Staff can add internal notes visible only to team members, never sent to the customer.	**Must**	AT
FR-CONV-016	Staff can close a conversation, moving it to `RESOLVED`.	**Must**	AT
FR-CONV-017	A conversation with no activity for 30 days (configurable) and no pending order auto-closes to `RESOLVED`.	**Should**	AT
FR-CONV-018	Staff can snooze a conversation until a chosen time.	**Could**	AT
FR-CONV-019	A conversation summary is generated automatically on entry to `HUMAN_REQUIRED` and on takeover, and regenerated if more than 10 new messages arrive before a human opens it.	**Must**	AT, EV
FR-CONV-020	Staff can see **why** the AI escalated, rendered from the stored decision trace and handoff context.	**Must**	MT

ID	Observable behaviour	Priority	Verify
FR-CONV-021	Concurrent processing of one conversation is prevented; two near-simultaneous inbound messages never produce interleaved or duplicated AI replies.	**Must**	AT
FR-CONV-022	Typing indicators and read receipts.	**Won't**	—
FR-CONV-030	The inbox presents three panes: conversation list (triage), chat thread, and customer context with actions.	**Must**	MT
FR-CONV-031	The conversation list is ordered by **sales priority**, not by recency: conversations in `HUMAN_REQUIRED` first (by handoff priority, then oldest wait), then AI-handled conversations by lead temperature, then lead score, then recency.	**Must**	AT
FR-CONV-032	The default view is "Needs me" (escalations plus hot leads), not "All conversations".	**Must**	MT
FR-CONV-033	Conversations are grouped and badged by lead temperature (hot / warm / cold).	**Must**	MT
FR-CONV-034	Each list card shows customer, last message snippet, temperature badge, product of interest, value, channel icon, assignee, and relative time — enough to triage without opening it.	**Must**	MT
FR-CONV-035	There is a single unified inbox across channels with channel filters; there are no per-channel pages.	**Must**	MT
FR-CONV-036	A channel appears in the filter list only when the business has a connected account on it; each conversation row displays its channel.	**Must**	AT
FR-CONV-037	The chat pane shows a persistent indicator of who is currently responding (AI or a named human) and renders customer, AI, human, and system events distinguishably.	**Must**	MT
FR-CONV-038	Outbound message delivery state (sending, sent, delivered, failed) is visible in the thread, and a failed provider send exposes a retry action.	**Must**	AT
FR-CONV-039	The customer panel shows identity, contact, lead information with an explanation affordance for the score, plain-language status, and an event timeline.	**Must**	MT
FR-CONV-040	The conversation timeline is derived from existing messages, state transitions, AI actions, and orders; it introduces no new storage.	**Must**	IN
FR-CONV-041	Primary actions (take over, confirm order, call, assign) are reachable from the customer panel, limited to four primary buttons, and context-sensitive to the conversation state.	**Must**	MT
FR-CONV-042	The call action opens a telephone link to the stored customer phone number.	**Should**	MT
FR-CONV-043	AI control (AI mode, take over, pause, resume, send) reflects **server** state, so two staff members can never both believe they own a conversation.	**Must**	AT
FR-CONV-044	No payment-link or payment-collection affordance exists anywhere in the UI while payments are out of scope.	**Must**	IN
FR-CONV-050	A pipeline page presents leads and orders as Kanban columns: new leads, qualified, payment pending, sold.	**Must**	MT
FR-CONV-051	Columns are a **read-only projection** of existing conversation and order status and introduce no new state or status field.	**Must**	IN
FR-CONV-052	Each column shows a count and total value; each card shows customer, product, value, and temperature.	**Must**	MT

ID	Observable behaviour	Priority	Verify
FR-CONV-053	Card copy for the payment-pending column says the order awaits **staff confirmation**, never that it awaits customer payment.	**Must**	IN
FR-CONV-054	Drag-to-advance between columns.	**Won't**	— (post-MVP; must execute state-machine transitions, never write status directly)
FR-LEAD-001	Every conversation carries a lead score in the range 0–100 and a derived temperature band.	**Must**	AT
FR-LEAD-002	The score is computed by deterministic rules over persisted signals; scoring must never invoke an LLM.	**Must**	AT, IN
FR-LEAD-003	The contributing signals and their point values are stored per conversation so the score can be explained in the UI.	**Must**	AT
FR-LEAD-004	The score is recomputed on inbound message, conversation state change, order creation and confirmation, and on a nightly decay pass.	**Must**	AT
FR-LEAD-005	Temperature thresholds are configurable per business rather than hard-coded.	**Must**	AT
FR-LEAD-006	The score resets when the conversation is resolved or its order is cancelled.	**Must**	AT
FR-LEAD-007	The score is never presented as a probability of purchase or any other statistical claim.	**Must**	IN
FR-LEAD-008	Lead score is visible to all staff roles, including Employees.	**Must**	AT
FR-LEAD-009	Lead scoring must never reorder a conversation ahead of one requiring human action; escalation ordering always wins.	**Must**	AT
FR-LEAD-010	Machine-learned or LLM-derived scoring.	**Won't**	—
FR-RT-001	New messages, status changes, handoffs, lead-score changes, and notifications reach an open UI without manual refresh.	**Must**	MT
FR-RT-002	Realtime connections authenticate with the same credentials as the REST API, and subscriptions are scoped server-side; a client may never subscribe to an arbitrary tenant or conversation.	**Must**	AT
FR-RT-003	The realtime channel is an accelerator, not a source of truth: no data may exist only in a pushed payload, and every view is refetchable over REST.	**Must**	IN
FR-RT-004	On reconnection the client reconciles by refetching the list and the open thread from its last cursor, so a missed frame never leaves the UI permanently wrong.	**Must**	AT
FR-RT-005	When the realtime connection cannot be established, the UI degrades to polling and indicates the degraded state.	**Must**	MT
FR-RT-006	Realtime fan-out works across multiple application instances.	**Must**	AT

10. FR-AI — Agent configuration and controlled autonomy

ID	Observable behaviour	Priority	Verify
FR-AI-001	Each business has exactly one configurable primary sales agent in MVP.	**Must**	AT
FR-AI-002	Owner/Admin can configure agent name, personality description, default language (`ar` / `fr` / `en`), tone, and greeting and fallback templates.	**Must**	AT
FR-AI-003	Owner/Admin can configure sales strategy flags (upsell aggressiveness, cross-sell on/off).	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-AI-004	Owner/Admin can set the escalation confidence threshold; default is 0.65.	**Must**	AT
FR-AI-005	Owner/Admin can configure model settings: primary model, fallback model, max tokens, temperature, retrieval top-k, blocked topics, and a custom system append.	**Should**	AT
FR-AI-006	Owner/Admin can enable or disable the agent globally for the business.	**Must**	AT
FR-AI-007	Configuration changes take effect on the next conversation turn without redeploy.	**Must**	AT
FR-AI-008	Multiple agents per business, per-channel agents.	**Won't**	— (Phase 5)
FR-AI-010	The business can operate in one of three reply modes: `auto_send`, `approval_required`, or `draft_only`.	**Must**	AT
FR-AI-011	New businesses default to `approval_required`. Completing the AI test gate (G4) unblocks live AI; it does **not** change `reply_mode`. The owner must explicitly opt into `auto_send` (onboarding step 6 optional toggle, or Settings).	**Must**	AT
FR-AI-012	In `approval_required`, AI output is persisted as a draft and is not delivered until a staff member approves it.	**Must**	AT
FR-AI-013	In `draft_only`, AI output is persisted as a draft that staff may copy or edit and send manually; approval does not auto-send.	**Should**	AT
FR-AI-014	Staff can edit a draft before approving; the edited text is what is sent, and the edit is recorded for feedback analysis.	**Should**	AT
FR-AI-015	Pending drafts are surfaced in the inbox as an actionable queue.	**Must**	MT
FR-AI-016	While a human is handling a conversation, the AI can offer a suggested reply that the human may send, edit, or dismiss.	**Must**	MT
FR-AI-017	A suggestion is a draft only: displaying one during `HUMAN_REQUIRED` does not violate the paused-AI rule, because nothing is delivered without an explicit human action.	**Must**	AT, IN
FR-AI-018	Suggestions obey every guardrail that applies to sent messages, including grounding, no discounts, no refund approval, and no payment promises.	**Must**	EV
FR-AI-019	Suggestions are suppressed on complaint and refund conversations.	**Must**	AT, EV
FR-AI-020a	Suggestions are generated on demand and debounced, never on every keystroke, and each generation is recorded in AI usage like any other model call.	**Must**	AT
FR-AI-020b	Sending a suggestion records an AI message flagged as human-approved; editing before sending records the edit for feedback analysis.	**Must**	AT
FR-AI-020	The AI must never state a price, stock quantity, delivery fee, discount, or policy that did not come from a tool result or business rule in the same turn.	**Must**	EV, AT
FR-AI-021	The AI must never invent products, promise unavailable items, grant unauthorised discounts, or approve refunds.	**Must**	EV
FR-AI-022	The AI replies in the business default language unless the customer clearly uses another supported language, in which case it matches the customer.	**Must**	EV
FR-AI-023	When tools return empty or conflicting data, the AI states uncertainty and offers human handoff rather than guessing.	**Must**	EV
FR-AI-024	On low classifier confidence the AI asks one clarifying question; a second consecutive low-confidence turn escalates.	**Must**	AT, EV
FR-AI-025	Every outbound AI message records intent, confidence, tools called, model, latency, and prompt version.	**Must**	AT
FR-AI-026	Prompts are versioned; the active version per type is immutable and a change creates a new version.	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-AI-027	Platform default prompts are seeded per business and can be overridden by the merchant.	**Must**	AT
FR-AI-028	If the primary model fails, the system retries with the fallback model before escalating.	**Must**	AT, OB
FR-AI-029	If all model attempts fail or exceed the latency ceiling, the conversation escalates to human with an integration/timeout reason rather than sending a degraded reply.	**Must**	AT
FR-AI-030	Owner/Admin can run a playground conversation against current configuration without touching a live channel.	**Must**	MT
FR-AI-031	The playground can run the four canned scenarios (price, angry customer, buying intent, recommendation) and free-form chat; it shows the AI response and tools called; the decision trace is visible to Owner/Admin only.	**Must**	MT
FR-AI-032	Sampled, golden-set, and staff-reviewed conversations persist an `ai_evaluations` row (`conversation_id`, `score`, `failure_reason`, `evaluator_type`, `created_at`) covering correctness, grounding, language, tools, escalation, and conversion.	**Must**	AT

11. FR-HAND — Human handoff

ID	Observable behaviour	Priority	Verify
FR-HAND-001	All fifteen defined `HUMAN_REQUIRED` triggers are implemented in MVP.	**Must**	AT, EV
FR-HAND-002	Every handoff records an escalation reason and a priority.	**Must**	AT
FR-HAND-003	Every handoff persists a staff briefing containing customer identity, conversation summary, intent, products discussed, AI recommendation, reason, linked pending order and amount when applicable, and the required action.	**Must**	AT
FR-HAND-004	Entering `HUMAN_REQUIRED` sets `ai_enabled = false`; the AI produces no further autonomous replies on that conversation.	**Must**	AT
FR-HAND-005	Entering `HUMAN_REQUIRED` emits a notification to Owner and Admin (and assignee when set) with the correct priority.	**Must**	AT
FR-HAND-006	The time a human first opens or takes the conversation is recorded as `handoff_completed_at`.	**Must**	AT
FR-HAND-007	The AI never upsells on a complaint and never confirms a refund or payment.	**Must**	EV
FR-HAND-008	On detected purchase intent, the AI confirms product and variant, verifies stock, collects required fields one question per turn, creates a pending order, triggers handoff, and tells the customer a team member will confirm — with no payment promise.	**Must**	AT, EV
FR-HAND-009	Purchase handoff re-verifies stock immediately before order creation; insufficient stock produces an apology and alternatives instead of an order.	**Must**	AT
FR-HAND-010	Staff can confirm a pending order from the conversation view in one action.	**Must**	AT
FR-HAND-011	Email, SMS, WhatsApp, and push notification of handoffs.	**Won't**	— (Phase 4)

12. FR-ORD, FR-CUST, FR-NOTIF — Commerce records

ID	Observable behaviour	Priority	Verify
FR-ORD-001	The AI can create an order with status `PENDING_CONFIRMATION` and payment status `NOT_COMPLETED`.	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-ORD-002	An order stores a customer snapshot (name, phone, address), line items with variant, quantity and unit price, totals, currency, source conversation, and creator (AI or user).	**Must**	AT
FR-ORD-003	Staff can create an order manually.	**Must**	AT
FR-ORD-004	Staff can transition an order to `CONFIRMED`, `CANCELLED`, or `COMPLETED`; invalid transitions are rejected.	**Must**	AT
FR-ORD-005	Confirming an order records `order_confirmed_at` and advances the conversation to `ORDER_CONFIRMED`.	**Must**	AT
FR-ORD-006	Staff can mark payment status as completed when payment is collected offline.	**Must**	AT
FR-ORD-007	Order creation is idempotent by idempotency key: a retried request returns the original order rather than creating a second one.	**Must**	AT
FR-ORD-008	Cancelling a pending order releases any associated stock reservation.	**Must**	AT
FR-ORD-009	The orders list is filterable by status and date and shows source (AI vs manual).	**Must**	MT
FR-ORD-010	Stock is reserved for a variant and quantity during order collection, with a default expiry of 30 minutes.	**Must**	AT
FR-ORD-011	Reservations move to committed on order creation, released on cancel or timeout, and expired reservations are swept by a scheduled job.	**Must**	AT
FR-ORD-012	Available stock exposed to the AI subtracts active reservations, preventing double-selling the last unit across concurrent conversations.	**Must**	AT
FR-ORD-013	Payment provider integration, in-chat checkout, payment links.	**Won't**	— (Phase 4)
FR-ORD-014	Fulfilment statuses (`PROCESSING`, `SHIPPED`, `DELIVERED`) and logistics integration.	**Won't**	—
FR-CUST-001	A customer record is created or matched on first inbound message via `customer_identities` (`platform`, `platform_user_id`). Same platform id is always the same customer.	**Must**	AT
FR-CUST-002	A customer stores name, phone, email, address, language, preferences, tags, and notes.	**Must**	AT
FR-CUST-003	Information collected by the AI during a conversation updates the customer record.	**Must**	AT
FR-CUST-004	Staff can view and edit a customer profile and see full purchase and conversation history.	**Must**	AT
FR-CUST-005	Staff can add tags and internal notes to a customer.	**Must**	AT
FR-CUST-006	Each customer has a lifecycle stage derived automatically from order and conversation events.	**Must**	AT
FR-CUST-007	Lifecycle stage influences AI tone and recommendation behaviour as specified in `05_AI_BEHAVIOR_SPECIFICATION.md`.	**Must**	EV
FR-CUST-008	A customer can be manually tagged VIP; VIP handling is governed by the handoff rules.	**Must**	AT
FR-CUST-009	Owner/Admin can export customer data.	**Should**	AT
FR-CUST-010	Owner/Admin can manually merge two customer records; the merge is audited; Employee cannot merge.	**Must**	AT
FR-CUST-011	A new identity with a **verified** phone that already belongs to a customer on the same business is auto-linked. Unverified phone or name similarity never auto-merges.	**Must**	AT
FR-NOTIF-001	The system creates an in-app notification for new order requests and for handoffs, carrying type, priority, title, message, and linked entity.	**Must**	AT
FR-NOTIF-002	Notifications are targeted at Owner and Admin, plus the assignee when a conversation is assigned.	**Must**	AT
FR-NOTIF-003	The UI shows an unread count and a notification list sorted by priority then recency.	**Must**	MT

ID	Observable behaviour	Priority	Verify
FR-NOTIF-004	A notification links directly to the relevant conversation or order.	**Must**	MT
FR-NOTIF-005	Users can mark one notification or all notifications as read.	**Must**	AT
FR-NOTIF-006	New notifications appear without page refresh.	**Should**	MT
FR-NOTIF-007	Email, SMS, WhatsApp, browser push channels.	**Won't**	— (Phase 4)
FR-NOTIF-010	Notifications fire on exactly four situations: a new hot lead, an order request or payment confirmation, AI uncertainty (low confidence, policy conflict, integration failure), and an angry or complaining customer.	**Must**	AT
FR-NOTIF-011	Routine AI replies, routine inbound messages, and state changes with no human action attached never produce a notification.	**Must**	AT
FR-NOTIF-012	Notifications are deduplicated to at most one per conversation per trigger type per hour.	**Must**	AT
FR-NOTIF-013	Repeated hot-lead notifications within a short window are collapsed into a single grouped notification.	**Must**	AT
FR-NOTIF-014	Only `critical` notifications interrupt; lower priorities accumulate in the notification centre.	**Must**	MT
FR-NOTIF-015	Non-critical notifications respect the business's configured working hours.	**Should**	AT

13. FR-ANL, FR-BILL, FR-OBS, FR-FLAG, FR-AUD — Platform surfaces

ID	Observable behaviour	Priority	Verify
FR-ANL-001	The dashboard shows exactly the four KPIs defined in §15 for a selectable period.	**Must**	AT
FR-ANL-002	Handoffs are broken down by escalation reason.	**Must**	AT
FR-ANL-003	KPI values are computed from daily rollups, not live table scans, and are consistent for the same period across requests.	**Must**	AT
FR-ANL-004	Employees see analytics read-only; Owner/Admin see the same figures.	**Must**	AT
FR-ANL-005	Custom reports, segmentation, exports, revenue charts, CSAT, cost per order, prompt A/B analysis.	**Won't**	— (behind feature_flags.advanced_analytics)
FR-BILL-001	Plans define limits for channel accounts, AI-handled messages per month, max conversations, max AI replies, seats, and an optional monthly AI budget.	**Must**	AT
FR-BILL-002	Each business has a subscription with plan, status, and period end.	**Must**	AT
FR-BILL-003	Usage events are recorded append-only for inbound messages, AI outbound messages, and orders created.	**Must**	AT
FR-BILL-004	Usage is aggregated per business per period on a schedule and used for limit enforcement.	**Must**	AT
FR-BILL-005	The owner sees current usage against plan limits.	**Must**	MT
FR-BILL-006	Soft warnings are surfaced at 80% and 95% of the message limit.	**Must**	AT
FR-BILL-007	On reaching any hard usage limit (messages, AI replies, conversations, or monthly AI budget), AI auto-reply stops, inbound messages still arrive in the inbox, humans can still reply, the owner is notified, and an upgrade CTA is shown.	**Must**	AT

ID	Observable behaviour	Priority	Verify
FR-BILL-008	Exceeding the channel limit blocks connecting a further channel account.	**Must**	AT
FR-BILL-009	Self-service plan upgrade and payment collection.	**Should**	AT
FR-BILL-010	The system records `ai_tokens`, `ai_cost`, and `messages_processed` and uses them for limit enforcement and Owner cost visibility.	**Must**	AT
FR-OBS-001	Every tool call is logged with tool name, input, output, success flag, error, and duration.	**Must**	AT
FR-OBS-002	Every AI turn stores a decision trace containing intent, confidence, rules checked, policy outcomes, decision, escalation reason, priority, and prompt version.	**Must**	AT
FR-OBS-003	Owner and Admin can view the raw decision trace for a conversation (including playground). Employees see the handoff briefing (reason, priority, required action), not the reasoning trace.	**Must**	MT
FR-OBS-004	Token counts and estimated cost are recorded per AI turn with model and prompt version.	**Must**	AT
FR-OBS-005	Owner can view an AI usage and cost summary.	**Should**	MT
FR-OBS-006	Staff can rate any AI message positive or negative, with a reason from a fixed enum and an optional comment.	**Must**	AT
FR-OBS-007	Feedback is filterable by date, reason, and prompt version for quality review.	**Must**	AT
FR-OBS-008	Rate-limit events are logged as AI actions.	**Must**	AT
FR-OBS-009	AI action logs are retained 90 days in the primary database with an archival path defined.	**Should**	IN
FR-OBS-010	Automated fine-tuning from feedback.	**Won't**	—
FR-OBS-011	Conversation-level `ai_evaluations` are persisted as specified in FR-AI-032.	**Must**	AT
FR-FLAG-001	Feature flags are stored per business with a unique feature key and enabled state.	**Must**	AT
FR-FLAG-002	Platform-level flags (no business scope) provide defaults and beta cohort overrides.	**Must**	AT
FR-FLAG-003	Flags are checked by the integrations UI, the AI orchestrator, and billing limit logic.	**Must**	IN
FR-FLAG-004	Flag changes take effect without redeploy.	**Must**	AT
FR-AUD-001	Security- and money-relevant actions by users are recorded in an audit log with actor, action, entity, and payload.	**Must**	AT
FR-AUD-002	Owner and Admin can view the audit log for their business.	**Should**	MT
FR-AUD-003	AI actions are logged separately from human audit logs and are never conflated.	**Must**	IN
FR-AUD-004	Raw provider webhook payloads are retained for replay and debugging.	**Must**	AT

Activation gates

The agent cannot go live until every gate is true. The API rejects activation when any gate fails (FR-BIZ-024, FR-BIZ-025). The value is computed; it is not a stored column.

Gate	Condition
G1 Catalog	At least one active product with a variant that has a price and a stock value.
G2 Rules	Delivery, payment method, and returns are present. Accepting the seeded defaults counts.
G3 Channel	At least one channel account is active.
G4 AI test	A playground exchange, or one approved reply during an approval trial. This does not change reply mode.
G5 Handoff	The owner has acknowledged that purchase intent becomes a pending order for a human.

Traceability

Requirements	Stories	Primary design
FR-AUTH, FR-BIZ	Epic A	Security design and the onboarding activity
FR-CAT, FR-KNOW	Epic B	Identity and catalog class diagram
FR-INT	Epic C	Channel-connect sequence and deployment diagram
FR-CONV, FR-AI, FR-LEAD, FR-RT	Epic D	Inbound sequence, inbound activity, conversation state
FR-HAND, FR-ORD	Epic E	Purchase sequence and purchase activity
FR-CUST, FR-NOTIF	Epic F	Commerce class diagram and oversight use cases
FR-ANL, FR-BILL, FR-OBS, FR-FLAG, FR-AUD	Epic G	Architecture overview and oversight use cases